

SPT-Txn — CASP Travel Rule Compliance Kit

Privacy-preserving FATF Travel Rule for the MiCA era — embed it, don't outsource your customer data.

Violet Sky Security SEZC (Cayman) · Open source (Apache-2.0) · foss.violetskysecurity.com



The contradiction we resolve

A MiCA-authorised CASP must satisfy the EU Transfer of Funds Regulation (2023/1113) — originator and beneficiary information on every crypto-asset transfer — while honouring GDPR data-minimisation. Plain Travel Rule transports satisfy the first by shipping full PII to the counterparty, maximising what the other side holds. SPT-Txn resolves the tension cryptographically: the counterparty learns a transfer is reportable, between registered CASPs, with an attested identity — without receiving the amount or the identity fields it is not entitled to see.

What's in the kit

Component	What it does (TFR / MiCA relevance)
Embeddable verifier (pkg/verify)	Eight-step offline enforcement engine you run inside your infrastructure — SPT-Txn holds no data, so it is not a data processor and shrinks your third-party risk surface.
Privacy-preserving attestation	SD-JWT selective disclosure + three Groth16 zero-knowledge proofs: prove a transfer is reportable, between registered CASPs, with an attested identity — amount and counterparty stay hidden.
Self-hosted wallet proof-of-control	Signed-challenge verification for the TFR unhosted-wallet rule (transfers at or above €1000), cryptographically bound to the identity anchor.
Sunrise / fallback negotiation	Picks the strongest payload mode both CASPs support (ZK > sealed-TRISA > cleartext-TRP) and refuses anything below your security floor.
Missing-information policy engine	Configurable ACCEPT / HOLD / REJECT / REQUEST_MORE decisions — the TFR's missing-information procedure, as code.
Regulator disclosure	Minimal, verifiable package for a competent authority: reveal only the entitled fields + a Merkle inclusion proof of the audit record — data-minimisation by construction.
Transport & neutrality	TRP transport (live) + TRISA Secure-Envelope sealing core; chain-agnostic across 15 ledgers — the chain is an anchor, not a dependency.

Measured performance (reference POC, 10-core laptop — see docs/THROUGHPUT-BENCHMARK.md): **~62k** cleartext 3-hop authorization verifies/sec · **~394k** action-token mints/sec · ZK privacy path **~1.5 ms/verify (~6.7k/sec)**; proof generation ~159 ms, amortized once per delegation chain — not a per-action cost. Verification is stateless, so throughput scales with cores.

Why it's different

- ✓ Real zero-knowledge (Groth16/BN254, ~1.5 ms verify — measured) — not encrypt-in-transit.
- ✓ Embed, don't host — runs in your infra; we never hold your data.
- ✓ Multi-chain neutral — 15 ledgers; live testnet footprints (XRPL, Hedera, Sui, Starknet, EVM L2s).
- ✓ Open source (Apache-2.0) — fully auditable, no lock-in.

Honest status. Reference POC: open source, host-hardened (OpenBSD pledge/unveil), fuzzed (millions of executions, fails closed), security audit FAIL=0, conformance-vectored. The formal-assurance layer — an independent ZK/protocol audit, SOC 2, third-party pen test — is the funded next step. Engage as a design partner embedding the open-source library, where the architecture wins and the remaining items are stage-appropriate.

For chain ecosystems

- ✓ Give the CASPs & wallets on your chain a Travel Rule answer that meets MiCA + GDPR at once.
- ✓ Chain-agnostic by design — the ledger is an anchor, integrated not depended on.
- ✓ On-chain attestation anchors live across multiple testnets.
- ✓ A credible "Travel Rule is solved here" narrative for grants & BD.

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